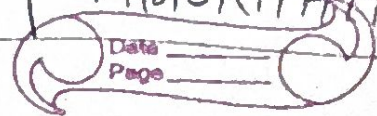


INTRO TO DATA STRUCTURE & ALGORITHMS



Problem 1: Algorithm properties and correctness

(a) Algorithm A

Algorithm A satisfies all five properties.

- ① Input $\rightarrow$ Takes temperature[] and n
- ② Output $\rightarrow$ Return average
- ③ Definiteness $\rightarrow$ clearly defined steps
- ④ finiteness $\rightarrow$ loop runs n time only
- ⑤ Effectiveness $\rightarrow$ Simple arithmetic operations.

Algorithm B

It has incorrect calculation logic so, it does not satisfy one property "effectiveness" and It produce incorrect output.

(b) Algorithm A total = 0

after each loop

$$\bullet i = 1 \rightarrow \text{total} = 20$$

$$\bullet i = 2 \rightarrow \text{total} = 45$$

$$\bullet i = 3 \rightarrow \text{total} = 67$$

$$\bullet i = 4 \rightarrow \text{total} = 91$$

$$\bullet i = 5 \rightarrow \text{total} = 112$$

$$\text{average} = \frac{112}{5} = 22.4 \checkmark$$

algorithm B sum = temperature[1] = 20

Add remaining

$$20 + 25 = 45$$

$$45 + 22 = 67$$

$$67 + 24 = 91$$

$$91 + 21 = 112$$

Now second loop divides repeatedly.

$$112/5 = 22.4$$

$$22.4/5 = 4.48$$

$$4.48/5 = 0.896$$

$$0.896/5 = 0.1792$$

$$0.1792/5 = 0.03584$$

incorrect output x

(c) logical error in Algorithm B

It divides n times instead of once.

correct average formula = $\frac{\sum \text{temperature}}{n}$

but B performs = $\frac{(((\text{sum})/n)/n)/n) \dots}{n}$

(d) Comparison

	Algo A	Algo B
Simplicity	Simple	confusing
correctness	correct	incorrect
efficiency	$O(n)$	$O(n)$ but wrong
Reliability	high	low

Best choice : Algo A (efficient, correct, simple)

(e) Test strategy

Test case	Input	Output
Normal case	[20, 25, 22, 24, 21]	22.4
Single value	[30], $n=1$	30
zero elements	[], $n=0$	error/undefined
Negative values	[-5, -10, -15]	-10
Mixed values	[10, -5, 15]	6.67

Problem 2 : Time complexity Analysis

(a) Worst case comparisons

Linear search

worst case $\rightarrow$ 10,000,000 comparisons.

Binary search

worst case $\rightarrow \log_2(10,000,000) = 23.25 \approx 24$

Hash Table

Worst case $\rightarrow$ 1 comparison

(b) Big-O Complexity

Algorithm	Complexity
Linear search	$O(n)$
Binary search	$O(\log n)$
Hash Table	$O(1)$

(c) Comparison table

n	Linear	Binary	hash
100	100	7	1
1000	1000	10	1
10,000	10,000	14	1
100,000	100,000	17	1
1,000,000	1,000,000	20	1
10,000,000	10,000,000	24	1

(d) Linear 10 ms $\times$ 50,000

500,000 ms

$\approx$ 500 seconds To slow

$$\begin{aligned}\text{binary} &= 24ns \times 50,000 \\ &= 1,200,000ns \\ &= 1.2ms \quad \text{Fast}\end{aligned}$$

$$\begin{aligned}\text{Hash} &= 1ns \times 50,000 \\ &= 50,000ns = 0.05ms \quad \text{faster}\end{aligned}$$

(c) Sorting v/s linear

sorting cost: $O(n \log n)$

binary search cost: $O(\log n) \times K$

linear cost: $K \times O(n)$

sorting beneficial when: $n \log n + K \log n < Kn$
when K is large, sorting is worth it

Problem-3 Space-time trade off

(a) implementation A

Nested loops: $O(n \times m)$

implementation B

hash creation: $O(n)$

checking prerequisites: $O(m)$ $= O(n+m)$

(b) space complexity

implementation

A

B

space

$O(1)$

$O(n)$

(c) Given

$n = 40$

$m = 5$

Students = 100000

Implementation A

$$\begin{aligned}\text{comparisons per student} &= 40 \times 5 = 200 \\ \text{total per day} &= 200 \times 100,000 = 20,000,000\end{aligned}$$

Implementation B

$$\begin{aligned}40 \times 8 &= 320 \text{ bytes} \\ 320 \times 100,000 &= 32,000,000 \text{ bytes} = 32 \text{ MB}\end{aligned}$$

(d) System constraint

Implementation A

$$\begin{aligned}200 \times 0.1 &= 20 \text{ microseconds} < 10s \\ \text{memory} &= \text{negligible.}\end{aligned}$$

Implementation B

$$\begin{aligned}320 \times 10,000 \\ 3,200,000 \text{ bytes} &= 3.2 \text{ MB} < 14 \text{ B}\end{aligned}$$

Implementation B is preferred because faster.

(e) hybrid approach

Threshold : $m = 4$

when $m \leq 4$ use nested loop (A)

when $m > 5$ use hash set (B)

Problem-4 Asymptotic notation

(a) bubble sort worst case

$$T(n) = n(n-1)/2$$

$$T(n) \leq n^2$$

$$T(n) = O(n^2)$$

Merge sort

$$T(n) = O(n \log n)$$

(b) Best algorithm

<u>Data type</u>	<u>Best algorithm</u>
Nearly sorted	Insertion sort
Random	Quick sort.
Reverse sorted	Merge sort

(c) Q v/s O

O means tight bound.

$$\text{merge sort} = O(n \log n)$$

because best, worst, avg are same.

Quick sort $O(n \log n)$

Average case only

$O(n^2) \rightarrow$ worst case.

(d) Decision tree

Small data $\rightarrow$ Insertion sort

Medium data $\rightarrow$ Quick sort

Large data $\rightarrow$ Merge sort

(e) Mystery sort

$$5n^2 + 100n + 1000 < 10n \log_2 n$$

for large n $5n^2 > 10n \log_2 n$

Since $n^2 > 2n \log_2 n$

• mystery sort becomes much slower for large n .